

NYERI, Kenya

WMO: 637170

Lat: 0.4421S

Lon: 36.9808E

Elev: 5900

StdP: 11.82

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	49.3	51.4	41.8	48.5	66.5	44.8	54.7	65.0	16.5	65.8	13.9	66.4	3.0	230	0.334

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	21.9	81.5	61.2	79.9	61.5	78.3	61.7	66.1	74.1	65.4	73.4	64.7	72.8	9.4	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.6	110.1	69.4	62.9	107.0	68.3	62.3	104.9	67.6	34.5	74.3	33.8	73.6	33.3	73.1	74.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
15.9	13.5	11.7	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			44.4	68.3	2.6	1.4	42.5	69.3	41.0	70.1	39.6	70.9	37.7	71.9		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	65.1	65.2	66.7	67.3	67.3	65.8	63.6	61.7	62.1	64.8	66.6	65.1	64.7
	DBStd	2.83	2.05	2.31	2.51	2.02	1.40	2.11	2.53	2.59	2.59	1.93	1.90	1.99
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	394	22	8	8	4	8	51	107	94	34	8	21	29
	CDD50	5498	472	468	536	520	490	408	361	375	444	514	453	457
	CDD65	419	29	57	79	74	34	9	3	4	28	57	24	21
	CDH74	2059	323	565	570	184	16	5	5	7	109	127	44	104
	CDH80	140	8	58	54	14	0	0	0	1	2	2	2	1
(14) Wind	WSAvg	5.0	4.7	4.8	5.1	5.0	4.9	4.8	4.5	5.2	6.0	5.4	4.6	4.7
(15) Precipitation	PrecAvg	44.4	3.1	2.4	3.5	7.4	6.6	1.8	1.7	2.2	1.5	5.1	5.9	3.5
	PrecMax	58.9	9.8	5.7	9.5	12.3	13.4	3.3	2.6	4.5	3.1	12.9	11.9	5.7
	PrecMin	31.3	1.4	1.0	0.9	4.3	3.2	0.6	0.9	1.1	0.8	0.9	3.5	1.1
	PrecStd	7.2	2.2	1.2	1.9	2.5	3.0	0.6	0.6	0.8	0.6	3.0	2.4	1.5
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	81.0	84.1	83.9	80.8	75.6	75.0	74.6	74.8	79.5	79.3	77.2	78.8
		MCWB	59.8	59.4	61.0	63.7	64.7	63.1	61.6	60.9	61.2	61.5	63.1	61.1
	2%	DB	79.2	81.7	81.5	77.8	74.3	72.5	72.0	72.5	77.0	77.1	75.3	76.6
		MCWB	60.9	60.4	61.9	63.5	64.3	62.8	60.9	60.8	60.9	61.5	62.7	62.0
	5%	DB	77.5	79.8	79.6	75.8	73.1	70.9	70.0	70.5	75.1	75.3	73.6	74.8
		MCWB	61.3	61.1	62.0	63.5	63.9	62.2	60.4	60.3	60.8	61.7	62.7	62.1
	10%	DB	75.8	77.6	77.5	74.1	71.7	69.4	67.8	68.4	73.0	73.5	71.9	73.1
		MCWB	61.6	61.6	62.1	63.5	63.4	61.5	59.8	59.7	60.4	61.4	62.4	62.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	66.3	66.6	66.6	67.0	66.4	65.3	63.9	63.7	64.2	65.0	65.5	66.4
		MCDB	74.9	74.4	76.5	75.0	73.2	72.1	71.7	71.8	74.8	73.7	72.9	73.2
	2%	WB	64.8	65.3	65.4	65.7	65.2	63.8	62.3	62.2	62.9	63.9	64.5	64.8
		MCDB	73.5	74.5	75.3	73.3	72.1	70.4	69.5	69.8	73.0	72.1	71.4	71.8
	5%	WB	63.9	64.2	64.5	64.8	64.5	62.9	61.2	61.1	62.0	63.2	63.7	64.0
		MCDB	72.6	74.4	74.3	72.4	71.4	69.0	67.8	68.3	71.7	71.1	70.4	70.8
	10%	WB	63.0	63.2	63.7	64.1	63.8	62.0	60.2	60.1	61.1	62.5	63.0	63.2
		MCDB	71.5	73.3	73.0	71.5	70.1	67.6	66.0	66.5	70.3	70.2	69.4	69.7
(35) Mean Daily Temperature Range	5% DB	MDBR	23.1	24.8	21.9	15.5	13.0	12.4	12.3	13.3	16.5	15.9	15.2	18.9
		MCDBR	26.5	28.7	26.2	18.2	14.6	14.7	16.5	17.0	20.2	18.3	17.9	22.2
		MCWBR	11.7	11.8	10.4	6.8	6.1	6.4	7.4	7.3	7.4	6.1	7.5	10.3
	5% WB	MCDBR	21.8	22.8	20.2	15.6	13.6	13.6	14.6	15.3	17.5	15.7	15.6	17.9
		MCWBR	11.1	10.4	9.0	6.6	6.2	6.6	7.3	7.3	7.4	6.2	7.2	9.2
(40) Clear-Sky Solar Irradiance	taub		0.356	0.378	0.373	0.331	0.329	0.388	0.405	0.378	0.350	0.352	0.347	0.353
	taud		2.470	2.388	2.433	2.610	2.605	2.387	2.313	2.393	2.469	2.482	2.522	2.498
	Ebn at Noon		309	302	301	307	300	278	275	289	303	307	309	308
	Edh at Noon		36	40	38	31	30	37	40	38	36	36	35	35
(44) All-Sky Solar Radiation	RadAvg		1965	2075	2008	1781	1719	1528	1440	1524	1823	1822	1668	1787
	RadStd		153	115	152	113	65	101	133	100	95	88	101	128

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)	+0.85	+1.24	N/A	+1.03	+0.70	+0.29	N/A	N/A	N/A	N/A

Nomenclature: See separate page